SUMS OF POWERS VIA CENTRAL FINITE DIFFERENCES AND NEWTON'S FORMULA

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Abstract. In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further provide Wolfram Mathematica programs for the efficient verification of the derived identities.

Contents

1. Auxiliary Lemmas 2
2. Introduction 5
3. Main Results 6
4. Remark on Knuth's Formula 11
5. Conclusions 16
References 16
Mathematica Programs 18

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1. Auxiliary Lemmas

This section introduces the lemmas, definitions, and conventions used in the main results. We use Donald Knuth's notation for *multifold sums of powers* [1] because it is concise.

Definition 1.1 (Multifold sums of powers)**.** *For non-negative integers r, n, and m, define the* multifold sums of powers *recursively by*

$$\begin{aligned}\Sigma^0\, n^m &= n^m,\\ \Sigma^1\, n^m &= \Sigma^0\, 1^m + \Sigma^0\, 2^m + \cdots + \Sigma^0\, n^m,\\ \Sigma^{r+1}\, n^m &= \Sigma^r\, 1^m + \Sigma^r\, 2^m + \cdots + \Sigma^r\, n^m.\end{aligned}$$

We define the falling factorial notation as follows.

Definition 1.2 (Falling factorials)**.** *For integers n and k, define the* falling factorial *by*

$$(n)_k = \begin{cases} 1, & \text{if } k = 0,\\ n(n-1)(n-2)\cdots(n-k+1) = \prod_{j=0}^{k-1}(n-j), & \text{if } k > 0,\\ 0, & \text{otherwise.}\end{cases}.$$

We define the central factorial notation as follows.

Definition 1.3 (Central factorials)**.** *For integers n and k, define the* central factorial *by*

$$n^{[k]} = \begin{cases} 0, & \text{if } k < 0,\\ 1, & \text{if } k = 0,\\ n\left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = n\left(n+\frac{k}{2}-1\right)_{k-1}, & \text{if } k > 0,\end{cases},$$

where $\left(n+\frac{k}{2}-1\right)_{k-1} = \prod_{j=0}^{k-2}\left(n+\frac{k}{2}-1-j\right)$ is a falling factorial.

For further discussion of generalized factorials, see J. F. Steffensen's work [2].

Lemma 1.4 (Central Newton's formula)**.** *Let f be a function defined on the integers, and let x and t be arbitrary integers.*

$$f(x) = \sum_{k=0}^{\infty} \frac{(x-t)^{[k]}}{k!} \delta^k f(t).$$

The kth central finite difference is defined by

$$\delta^k f(t) = \sum_{j=0}^{k} (-1)^j \binom{k}{j} f\left(t + \tfrac{k}{2} - j\right),$$

and $(x-t)^{[k]}$ denotes the central factorial defined in Definition (1.3).

Proof. See [3, p. 462], [4, Section 2, Equation (19)], and [5, Section 6, Equation (21)]. □

The specialization to powers gives the following lemma.

Lemma 1.5 (Newton's formula for powers)**.** *Let n and m be nonnegative integers, and let t be an arbitrary integer.*

$$n^m = \sum_{k=0}^{m} \frac{(n-t)^{[k]}}{k!} \delta^k t^m.$$

Proof. This is the special case of Lemma (1.4) with $f(n) = n^m$. □

The central factorial identity has the following binomial form.

Lemma 1.6 (Central factorials binomial form)**.** *For integers n and nonzero k,*

$$\frac{n^{[k]}}{k!} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1}.$$

Proof. We have

$$\frac{n^{[k]}}{k!} = \frac{n}{k!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k(k-1)!} \left(n + \frac{k}{2} - 1\right)_{k-1} = \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1},$$

by the falling-factorial identity $\frac{(x)_n}{n!} = \binom{x}{n}$. □

We also use the generalized hockey-stick identity.

Lemma 1.7 (Generalized hockey-stick identity)**.** *For arbitrary integers a, b, and j,*

$$\sum_{k=a}^{b} \binom{k}{j} = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

Proof. We have $\sum_{k=a}^{b} \binom{k}{j} = \binom{a}{j} + \binom{a+1}{j} + \cdots + \binom{b}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right)$. The hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ therefore gives

$$\sum_{k=a}^{b} \binom{k}{j} = \left(\sum_{k=0}^{b} \binom{k}{j}\right) - \left(\sum_{k=0}^{a-1} \binom{k}{j}\right) = \binom{b+1}{j+1} - \binom{a}{j+1}.$$

This completes the proof. □

The centered version is as follows.

Lemma 1.8 (Centered hockey-stick identity)**.** *Let n be a positive integer, and let t, k, and r be arbitrary integers.*

$$\sum_{j=1}^{n} \binom{j - t + \frac{k}{2} + r}{k} = \binom{n - t + \frac{k}{2} + r + 1}{k+1} - \binom{1 - t + \frac{k}{2} + r}{k+1}.$$

Proof. Substituting $a = 1 - t + \frac{k}{2} + r$ and $b = n - t + \frac{k}{2} + r$ into Lemma (1.7) gives

$$\sum_{j=1}^{n} \binom{j-t+\frac{k}{2}+r}{k} = \sum_{j=1-t+\frac{k}{2}+r}^{n-t+\frac{k}{2}+r} \binom{j}{k} = \binom{n-t+\frac{k}{2}+r+1}{k+1} - \binom{1-t+\frac{k}{2}+r}{k+1}.$$

This completes the proof. □

The following formula gives the multifold sum of zeroth powers.

Lemma 1.9 (Multifold sums of zero powers)**.** *Let r and n be nonnegative integers.*

$$\Sigma^r\, n^0 = \binom{r+n-1}{r}.$$

Proof.

(1) When $r = 0$, Definition (1.1) gives $\Sigma^0\, n^0 = 1$.

(2) When $r = 1$, we have $\Sigma^1\, n^0 = \sum_{k=1}^{n} 1 = \binom{n}{1}$.

(3) When $r = 2$, we have $\Sigma^2\, n^0 = \sum_{k=1}^{n} \binom{k}{1} = \binom{n+1}{2}$.

(4) When $r = 3$, we have $\Sigma^3\, n^0 = \sum_{k=1}^{n} \binom{k+1}{2} = \binom{n+2}{3}$.

(5) For general r, the hockey-stick identity gives $\Sigma^r\, n^0 = \sum_{k=1}^{n} \Sigma^{r-1}\, k^0 = \sum_{k=1}^{n} \binom{k+r-2}{r-1} = \binom{r+n-1}{r}$.

This completes the proof. □

Throughout, we use the following convention.

Convention 1.10. *For all x,*

$$x^0 = 1.$$

Donald Knuth discusses the convention $x^0 = 1$ for all x, including zero, in Concrete Mathematics [6, p. 162] and in [7].

2. Introduction

In this manuscript, we derive formulas for sums of powers by combining the central Newton interpolation formula for powers in Lemma (1.5) with the hockey-stick identity in Lemma (1.8). We implement the algorithm for finding closed forms for sums of powers of integers proposed in [8, Alg. (11.2)]. More precisely, the main idea is to identify the binomial basis in variants of Newton's formula associated with different difference operators. For example,

$$\begin{cases} f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{[k]}}{k!} \delta^k f(a) = \sum_{k=0}^{\infty} \frac{x-a}{k} \binom{x-a+\frac{k}{2}-1}{k-1} \delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)_k}{k!} \Delta^k f(a) = \sum_{k=0}^{\infty} \binom{x-a}{k} \Delta^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^{(k)}}{k!} \nabla^k f(a) = \sum_{k=0}^{\infty} \binom{x-a+k-1}{k} \nabla^k f(a), \\ f(x) &= \sum_{k=0}^{\infty} \frac{(x-a)^k}{k!} \frac{\mathrm{d}^k f(a)}{\mathrm{d}x^k}. \end{cases}.$$

Each variant of Newton's formula above has a corresponding binomial basis:

$$\begin{cases} \frac{(x)_n}{n!} &= \frac{1}{n!} x(x-1)(x-2)\cdots(x-n+1) = \binom{x}{n}, \\ \frac{x^{(n)}}{n!} &= \frac{1}{n!} x(x+1)(x+2)\cdots(x+n-1) = \binom{x+n-1}{n}, \\ \frac{x^{[n]}}{n!} &= \frac{1}{n!} \left(n+\frac{k}{2}-1\right)\left(n+\frac{k}{2}-2\right)\cdots\left(n-\frac{k}{2}+1\right) = \frac{x}{n}\binom{x+\frac{n}{2}-1}{n-1}. \end{cases}.$$

These bases allow us to combine Newton's formula with a hockey-stick identity to obtain closed forms for sums of powers. Thus, we focus on closed forms for sums of powers using the central Newton formula and the hockey-stick identity in Lemma (1.8).

3. Main Results

We begin with Lemma (1.5), which gives Newton's interpolation formula for powers. Applying Lemma (1.6) to this formula, for non-negative integers n and m and an arbitrary integer t, gives

$$\begin{aligned} n^m &= \frac{(n-t)^{[0]}}{0!}\delta^0 t^m + \sum_{k=1}^{m} \frac{n-t}{k}\binom{n+t+\frac{k}{2}-1}{k-1}\delta^k t^m \\ &= t^m + \sum_{k=1}^{m}(n-t)\binom{n-t+\frac{k}{2}-1}{k-1}\frac{\delta^k t^m}{k}. \end{aligned}$$

Note that we isolate the term $\frac{(n-t)^{[0]}}{0!}\delta^0 t^m$ to avoid division by zero. By expanding the brackets, we have

$$n^m = t^m + \sum_{k=1}^{m} \frac{\delta^k t^m}{k}\left[n\binom{n-t+\frac{k}{2}-1}{k-1} - t\binom{n-t+\frac{k}{2}-1}{k-1}\right].$$

The sum above contains the additional factor n in its binomial basis, $n\binom{n-t+\frac{k}{2}-1}{k-1}$, which prevents direct application of the centered hockey-stick identity in Lemma (1.8). We first establish the following lemma.

Lemma 3.1 (Binomial decomposition)**.** *Let r, n, and m be nonnegative integers.*

$$n\binom{n+r}{m} = (m+1)\binom{n+r}{m+1} - (r-m)\binom{n+r}{m}.$$

Proof. Expanding the right-hand side gives

$$n\binom{n+r}{m} = m\binom{n+r}{m+1} + \binom{n+r}{m+1} - r\binom{n+r}{m} + m\binom{n+r}{m}.$$

Recall the extraction property of binomial coefficients $\binom{n}{k+1} = \frac{n-k}{k+1}\binom{n}{k}$. The extraction property then gives

$$\binom{n+r}{m+1} = \frac{n+r-m}{m+1}\binom{n+r}{m}.$$

Thus,

$$n\binom{n+r}{m} = m\frac{n+r-m}{m+1}\binom{n+r}{m} + \frac{n+r-m}{m+1}\binom{n+r}{m} - r\binom{n+r}{m} + m\binom{n+r}{m}.$$

Factoring out the binomial coefficient $\binom{n+r}{m}$ gives

$$\begin{aligned} n\binom{n+r}{m} &= \binom{n+r}{m}\left[m\frac{n+r-m}{m+1} + \frac{n+r-m}{m+1} - r + m\right] \\ &= \binom{n+r}{m}\left[(m+1)\frac{n+r-m}{m+1} - r + m\right] \\ &= \binom{n+r}{m}n. \end{aligned}$$

This completes the proof. □

Substituting $r = -t + \frac{k}{2} - 1$ and $m = k-1$ into Lemma (3.1) gives

$$n\binom{n-t+\frac{k}{2}-1}{k-1} = k\binom{n-t+\frac{k}{2}-1}{k} + \left[t+\frac{k}{2}\right]\binom{n-t+\frac{k}{2}-1}{k-1}.$$

For clarity, applying Lemma (3.1) directly gives

$$\begin{aligned} n\tbinom{n-t+\frac{k}{2}-1}{k-1} &= k\tbinom{n-t+\frac{k}{2}-1}{k-1+1} - \left[-t+\tfrac{k}{2}-1-(k-1)\right]\tbinom{n-t+\frac{k}{2}-1}{k-1} \\ &= k\tbinom{n-t+\frac{k}{2}-1}{k} - \left[-t-\tfrac{k}{2}\right]\tbinom{n-t+\frac{k}{2}-1}{k-1} \\ &= k\tbinom{n-t+\frac{k}{2}-1}{k} + \left[t+\tfrac{k}{2}\right]\tbinom{n-t+\frac{k}{2}-1}{k-1}. \end{aligned}$$

Thus,

$$n^m = t^m + \sum_{k=1}^{m}\frac{\delta^k t^m}{k}\left[k\binom{n-t+\frac{k}{2}-1}{k} + \left[t+\frac{k}{2}\right]\binom{n-t+\frac{k}{2}-1}{k-1} - t\binom{n-t+\frac{k}{2}-1}{k-1}\right].$$

Combining the terms involving $t\binom{n-t+\frac{k}{2}-1}{k-1}$ and simplifying the factors of k gives

$$n^m = t^m + \sum_{k=1}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}\right]\delta^k t^m.$$

We absorb the term t^m into the sum to obtain

$$n^m = \sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}\right]\delta^k t^m.$$

We now simplify the binomial basis $\binom{n-t+\frac{k}{2}-1}{k} + \frac{1}{2}\binom{n-t+\frac{k}{2}-1}{k-1}$ to remove the fractional coefficient. Set $a = n - t + \frac{k}{2} - 1$.

$$\binom{a}{k} + \frac{1}{2}\binom{a}{k-1} = \frac{1}{2}\left[2\binom{a}{k} + \binom{a}{k-1}\right] = \frac{1}{2}\left[\binom{a}{k} + \binom{a+1}{k}\right].$$

Therefore,

$$n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}-1}{k} + \binom{n-t+\frac{k}{2}}{k-1}\right]\delta^k t^m.$$

The upper indices of the binomial coefficients may be fractional, because $\binom{n}{k}$ is a polynomial in n. The identity involving the unsigned Stirling numbers of the first kind $\left[\begin{smallmatrix} n \\ k \end{smallmatrix}\right]$ makes this explicit:

$$\binom{n}{k} = \sum_{j=0}^{k}\frac{(-1)^{k-j}}{k!}\left[\begin{smallmatrix} k \\ j \end{smallmatrix}\right] n^j.$$

The following example illustrates the identity.

Example 3.2. *For integer* $0 \le t \le 3$*, we have*

$$\begin{aligned}
t = 0: \quad n^3 &= \left(1 + \tbinom{n}{-1}\right)\cdot 0 + \tfrac{1}{2}(1+2n)\cdot\tfrac{1}{4} + \tfrac{1}{2}(2+n+n^2)\cdot 0 \\
&\quad + \tfrac{1}{48}(21 + 46n + 12n^2 + 8n^3)\cdot 6, \\
t = 1: \quad n^3 &= \left(1 + \tbinom{n-1}{-1}\right)\cdot 1 + \tfrac{1}{2}(-1+2n)\cdot\tfrac{13}{4} + \tfrac{1}{2}(2-n+n^2)\cdot 6 \\
&\quad + \tfrac{1}{48}(-21 + 46n - 12n^2 + 8n^3)\cdot 6, \\
t = 2: \quad n^3 &= \left(1 + \tbinom{n-2}{-1}\right)\cdot 8 + \tfrac{1}{2}(-3+2n)\cdot\tfrac{49}{4} + \tfrac{1}{2}(4-3n+n^2)\cdot 12 \\
&\quad + \tfrac{1}{48}(-87 + 94n - 36n^2 + 8n^3)\cdot 6, \\
t = 3: \quad n^3 &= \left(1 + \tbinom{n-3}{-1}\right)\cdot 27 + \tfrac{1}{2}(-5+2n)\cdot\tfrac{109}{4} + \tfrac{1}{2}(8-5n+n^2)\cdot 18 \\
&\quad + \tfrac{1}{48}(-225 + 190n - 60n^2 + 8n^3)\cdot 6.
\end{aligned}$$

We can now apply the centered hockey-stick identity in Lemma (1.8) to obtain a closed formula for sums of powers:

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}-1}{k}+\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}}{k-1}\right]\delta^k t^m.$$

Applying Lemma (1.8) gives

$$\begin{cases}\sum_{j=1}^{n}\binom{j-t+\frac{k}{2}-1}{k} &= \binom{n-t+\frac{k}{2}+0}{k+1}-\binom{0-t+\frac{k}{2}}{k+1},\\ \sum_{j=1}^{n}\binom{j-t+\frac{k}{2}}{k-1} &= \binom{n-t+\frac{k}{2}+1}{k+1}-\binom{1-t+\frac{k}{2}}{k+1}.\end{cases}.$$

We obtain the following closed formula for ordinary sums of powers.

Proposition 3.3 (Ordinary sums of powers)**.** *Let n and m be nonnegative integers, and let t be an arbitrary integer.*

$$\Sigma^1 n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+0}{k+1}+\binom{n-t+\frac{k}{2}+1}{k+1}-\binom{0-t+\frac{k}{2}}{k+1}-\binom{1-t+\frac{k}{2}}{k+1}\right]\delta^k t^m.$$

The following example illustrates the formula.

Example 3.4. *For integer $0 \le t \le 4$, we have*

$$\begin{aligned}
t=0: \quad \Sigma^1 n^3 &= 2n\cdot 0+(n+n^2)\cdot\tfrac{1}{4}+\tfrac{1}{6}(n+3n^2+2n^3)\cdot 0\\
&\quad+\tfrac{1}{24}(-n+n^2+4n^3+2n^4)\cdot 6,\\
t=1: \quad \Sigma^1 n^3 &= 2n\cdot 1+(-n+n^2)\cdot\tfrac{13}{4}+\tfrac{1}{6}(n-3n^2+2n^3)\cdot 6\\
&\quad+\tfrac{1}{24}(n+n^2-4n^3+2n^4)\cdot 6,\\
t=2: \quad \Sigma^1 n^3 &= 2n\cdot 8+(-3n+n^2)\cdot\tfrac{49}{4}+\tfrac{1}{6}(13n-9n^2+2n^3)\cdot 12\\
&\quad+\tfrac{1}{24}(-21n+25n^2-12n^3+2n^4)\cdot 6,\\
t=3: \quad \Sigma^1 n^3 &= 2n\cdot 27+(-5n+n^2)\cdot\tfrac{109}{4}+\tfrac{1}{6}(37n-15n^2+2n^3)\cdot 18\\
&\quad+\tfrac{1}{24}(-115n+73n^2-20n^3+2n^4)\cdot 6,\\
t=4: \quad \Sigma^1 n^3 &= 2n\cdot 64+(-7n+n^2)\cdot\tfrac{193}{4}+\tfrac{1}{6}(73n-21n^2+2n^3)\cdot 24\\
&\quad+\tfrac{1}{24}(-329n+145n^2-28n^3+2n^4)\cdot 6.
\end{aligned}$$

Similarly, setting $r = 0$ and $r = 1$ in Lemma (1.8) gives

$$\begin{cases} \sum_{j=1}^{n} \binom{j-t+\frac{k}{2}+0}{k+1} &= \binom{n-t+\frac{k}{2}+1}{k+2} - \binom{1-t+\frac{k}{2}}{k+2}, \\ \sum_{j=1}^{n} \binom{j-t+\frac{k}{2}+1}{k+1} &= \binom{n-t+\frac{k}{2}+2}{k+2} - \binom{2-t+\frac{k}{2}}{k+2}. \end{cases}$$

We can now state the formula for double sums of powers.

Proposition 3.5 (Double sums of powers)**.** *Let n and m be nonnegative integers, and let t be an arbitrary integer.*

$$\begin{aligned} \Sigma^2 n^m = \frac{1}{2} \sum_{k=0}^{m} \Bigg[& \binom{n-t+\frac{k}{2}+1}{k+2} + \binom{n-t+\frac{k}{2}+2}{k+2} \\ & - \left(\binom{0-t+\frac{k}{2}}{k+1} + \binom{1-t+\frac{k}{2}}{k+1} \right) \Sigma^1 n^0 \\ & - \left(\binom{1-t+\frac{k}{2}}{k+2} + \binom{2-t+\frac{k}{2}}{k+2} \right) \Sigma^0 n^0 \Bigg] \delta^k t^m. \end{aligned}$$

The following example illustrates the double-sum formula.

Example 3.6. *For integer $0 \le t \le 2$, we have*

$$\begin{aligned} t = 0: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(n+n^2) \cdot 0 + \tfrac{1}{48}(-3-2n+12n^2+8n^3) \cdot \tfrac{1}{4} \\ &\quad + \tfrac{1}{24}(-2n-n^2+2n^3+n^4) \cdot 0 \\ &\quad + \tfrac{1}{3840}(45+18n-200n^2-80n^3+80n^4+32n^5) \cdot 6, \\ t = 1: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(-n+n^2) \cdot 1 + \tfrac{1}{48}(3-2n-12n^2+8n^3) \cdot \tfrac{13}{4} \\ &\quad + \tfrac{1}{24}(2n-n^2-2n^3+n^4) \cdot 6 \\ &\quad + \tfrac{1}{3840}(-45+18n+200n^2-80n^3-80n^4+32n^5) \cdot 6, \\ t = 2: \quad \Sigma^1 n^3 &= \tfrac{1}{2}(2-3n+n^2) \cdot 8 + \tfrac{1}{48}(-15+46n-36n^2+8n^3) \cdot \tfrac{49}{4} \\ &\quad + \tfrac{1}{24}(-6n+11n^2-6n^3+n^4) \cdot 12 \\ &\quad + \tfrac{1}{3840}(105-142n-360n^2+560n^3-240n^4+32n^5) \cdot 6. \end{aligned}$$

The general case is as follows.

Theorem 3.7 (Multifold sums of powers)**.** *Let r, n, and m be nonnegative integers, and let t be an arbitrary integer.*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+r-1}{k+r} + \binom{n-t+\frac{k}{2}+r}{k+r} - \sum_{s=0}^{r-1}\left(\binom{s-t+\frac{k}{2}}{k+s+1} + \binom{s-t+\frac{k}{2}+1}{k+s+1}\right)\Sigma^{r-1-s} n^0\right]\delta^k t^m.$$

Lemma (1.9) gives the following formula in binomial form.

Proposition 3.8 (Multifold Binomial sums of powers)**.** *Let r, n, and m be nonnegative integers, and let t be an arbitrary integer.*

$$\Sigma^r n^m = \frac{1}{2}\sum_{k=0}^{m}\left[\binom{n-t+\frac{k}{2}+r-1}{k+r} + \binom{n-t+\frac{k}{2}+r}{k+r} - \sum_{s=0}^{r-1}\left(\binom{s-t+\frac{k}{2}}{k+s+1} + \binom{s-t+\frac{k}{2}+1}{k+s+1}\right)\binom{r-s+n-2}{r-s-1}\right]\delta^k t^m.$$

We have therefore derived closed formulas for multifold sums of powers. The work [9] implements the algorithm proposed in [8, Alg. (11.2)] using forward finite differences Δ and Newton's formula. Similarly, [10] discusses multifold formulas for sums of powers using backward differences ∇ and Newton's formula.

4. Remark on Knuth's Formula

Donald Knuth gives a formula for sums of odd powers in terms of the central factorial numbers of the second kind $T(n,k)$ [1].

Proposition 4.1 (Knuth's odd powers sum)**.** *Let n be a nonnegative integer, and let m be a positive integer.*

$$\Sigma^1 n^{2m-1} = \sum_{k=1}^{m}(2k-1)!\binom{n+k}{2k}T(2m,2k),$$

where $T(2m,2k)$ are central factorial numbers of the second kind.

The sequences A008957, A036969, and A269945 in the OEIS [11] list central factorial numbers of the second kind $T(n,k)$. Table 1 lists initial values of $T(2n,2k)$.

n/k	0	1	2	3	4	5	6	7
0	1							
1	0	1						
2	0	1	1					
3	0	1	5	1				
4	0	1	21	14	1			
5	0	1	85	147	30	1		
6	0	1	341	1408	627	55	1	
7	0	1	1365	13013	11440	2002	91	1

Table 1. Central factorial numbers of the second kind $T(2n,2k)$.

Evaluating Lemma (1.5) at zero yields Proposition (4.1). For $m \geq 1$, the formula gives

$$n^{2m} = \sum_{k=1}^{2m} \frac{n^{[k]}}{k!} \cdot \delta^k 0^{2m}.$$

The term for $k = 0$ vanishes because $\delta^0 0^{2m} = 0$ when $m \geq 1$. For positive integers m and k, the central difference $\delta^k 0^m$ is nonzero exactly when $k \leq m$ and m and k have the same parity:

$$\begin{cases} \delta^k 0^m \neq 0, & \text{when} \quad k \leq m \text{ and } m \equiv k \pmod 2, \\ \delta^k 0^m = 0, & \text{when} \quad m \not\equiv k \pmod 2, \end{cases}.$$

In particular, the difference vanishes when m and k have opposite parity. This parity property applies to powers evaluated at zero; for example, $\delta^3 1^4 = 24 \neq 0$. Since $2m$ is even, the differences $\delta^k 0^{2m}$ vanish for odd k, so we may omit those terms:

$$n^{2m} = \sum_{k=1}^{m} \frac{n^{[2k]}}{(2k)!} \cdot \delta^{2k} 0^{2m}.$$

By the definition of the central factorial numbers of the second kind, we obtain

$$n^{2m} = \sum_{k=1}^{m} n^{[2k]} T(2m, 2k).$$

Therefore,

$$n^{2m} = \sum_{k=1}^{m} (2k)! \frac{n^{[2k]}}{(2k)!} T(2m, 2k).$$

Applying Lemma (1.6) to simplify $\frac{n^{[2k]}}{(2k)!}$ gives

$$n^{2m} = \sum_{k=1}^{m} (2k)! \frac{n}{2k} \binom{n+k-1}{2k-1} T(2m, 2k).$$

Consequently,

$$n^{2m} = \sum_{k=1}^{m} (2k-1)! n \binom{n+k-1}{2k-1} T(2m, 2k).$$

Factoring out n gives

$$n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k-1}{2k-1} T(2m, 2k).$$

Applying the hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ gives

$$\Sigma^1 \, n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k}{2k} T(2m, 2k).$$

This formula agrees with Proposition (4.1). Repeated application of the hockey-stick identity $\sum_{k=0}^{n} \binom{k}{j} = \binom{n+1}{j+1}$ generalizes the formula to multifold sums.

Proposition 4.2 (Knuth's odd powers sum multifold)**.** *Let r and n be nonnegative integers, and let m be a positive integer.*

$$\Sigma^{r+1} \, n^{2m-1} = \sum_{k=1}^{m} (2k-1)! \binom{n+k+r}{2k+r} T(2m, 2k),$$

where $T(2m, 2k)$ are central factorial numbers of the second kind.

The following example illustrates Proposition (4.1).

Example 4.3. *For non-negative integers n, and $r = 1$, we have*

$$\Sigma^1\, n^1 = \tbinom{n+1}{2}1,$$

$$\Sigma^1\, n^3 = \tbinom{n+2}{4}6 + \tbinom{n+1}{2}1,$$

$$\Sigma^1\, n^5 = \tbinom{n+3}{6}120 + \tbinom{n+2}{4}30 + \tbinom{n+1}{2}1,$$

$$\Sigma^1\, n^7 = \tbinom{n+4}{8}5040 + \tbinom{n+3}{6}1680 + \tbinom{n+2}{4}126 + \tbinom{n+1}{2}1.$$

The following example illustrates Proposition (4.2).

Example 4.4. *For non-negative integers r, n*

$$\Sigma^r\, n^1 = \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r\, n^3 = \tbinom{n+1+r}{3+r}6 + \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r\, n^5 = \tbinom{n+2+r}{5+r}120 + \tbinom{n+1+r}{3+r}30 + \tbinom{n+0+r}{1+r}1,$$

$$\Sigma^r\, n^7 = \tbinom{n+3+r}{7+r}5040 + \tbinom{n+3+r}{5+r}1680 + \tbinom{n+1+r}{3+r}126 + \tbinom{n+0+r}{1+r}1.$$

The coefficients in Proposition (4.2) form the sequence A303675 in the OEIS [11]. Proposition (4.2) also motivates the following identity. Combining Newton's formula in Lemma (1.5) with Lemma (1.6) gives

$$n^m = \sum_{k=1}^{m} \frac{n}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^m.$$

Factoring out n gives

$$n^{m-1} = \sum_{k=1}^{m} \frac{1}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^m.$$

Shifting the exponent gives

$$n^m = \sum_{k=1}^{m+1} \frac{1}{k} \binom{n + \frac{k}{2} - 1}{k-1} \delta^k 0^{m+1}.$$

Re-indexing the sum gives

$$n^m = \sum_{k=0}^{m} \frac{1}{k+1} \binom{n + \frac{k}{2} - \frac{1}{2}}{k} \delta^{k+1} 0^{m+1}. \tag{1}$$

Setting $r = 0$ and $t = \frac{1}{2}$ in Lemma (1.8) gives the following proposition.

Proposition 4.5 (Ordinary sums of powers at zero)**.** *Let n and m be nonnegative integers.*

$$\Sigma^1 n^m = \sum_{k=0}^{m} \frac{1}{k+1} \left[\binom{n + \frac{k}{2} + \frac{1}{2}}{k+1} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \right] \delta^{k+1} 0^{m+1}.$$

The formula can also be written in terms of central factorial numbers of the second kind:

$$\Sigma^1 n^m = \sum_{k=0}^{m} k! \left[\binom{n + \frac{k}{2} + \frac{1}{2}}{k+1} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \right] T(k+1, m+1).$$

Setting $r = 1$ and $t = \frac{1}{2}$ in Lemma (1.8) gives a formula for double sums of powers.

Proposition 4.6 (Double sums of powers at zero)**.** *Let n and m be nonnegative integers.*

$$\Sigma^2 n^m = \sum_{k=0}^{m} \frac{1}{k+1} \left[\binom{n + \frac{k}{2} + \frac{3}{2}}{k+2} - \binom{\frac{k}{2} + \frac{1}{2}}{k+1} \Sigma^1 n^0 - \binom{\frac{k}{2} + \frac{3}{2}}{k+2} \Sigma^0 n^0 \right] \delta^{k+1} 0^{m+1}.$$

The general formula is as follows.

Theorem 4.7 (Multifold sums of powers at zero)**.** *Let r, n, and m be nonnegative integers.*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \Sigma^s n^0 \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

Proof. By repeatedly applied Lemma (1.8), and Equation (1). □

Combining Lemma (1.9) with Theorem (4.7) gives the following binomial form.

Proposition 4.8 (Multifold sums of powers at zero binomial form)**.** *Let r, n, and m be nonnegative integers.*

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] \frac{\delta^{k+1} 0^{m+1}}{k+1}.$$

Proof. By repeatedly applied Lemma (1.8), and Equation (1). □

This formula can also be expressed in terms of central factorial numbers of the second kind:

$$\Sigma^r n^m = \sum_{k=0}^{m} \left[\binom{n + \frac{k}{2} + \frac{2r-1}{2}}{k+r} - \sum_{s=0}^{r-1} \binom{\frac{k}{2} + \frac{2(r-s)-1}{2}}{k+r-s} \binom{s+n-1}{s} \right] k! T(k+1, m+1).$$

The results above follow from evaluating the central Newton formula at zero. Table 2 lists initial values of $T(n,k)$.

n/k	0	1	2	3	4	5	6	7	8
0	1								
1	0	1							
2	0	0	1						
3	0	$\frac{1}{4}$	0	1					
4	0	0	1	0	1				
5	0	$\frac{1}{16}$	0	$\frac{5}{2}$	0	1			
6	0	0	1	0	5	0	1		
7	0	$\frac{1}{64}$	0	$\frac{91}{16}$	0	$\frac{35}{4}$	0	1	
8	0	0	1	0	21	0	14	0	1

Table 2. Central factorial numbers of the second kind $T(n,k)$. See also the sequences A395862, and A370703 in the OEIS [11].

5. Conclusions

In this manuscript, we derive closed formulas for multifold sums of powers of integers by combining the central Newton interpolation formula with hockey-stick identities for binomial coefficients. We further obtain representations of multifold sums of powers in terms of Stirling numbers of the second kind and Eulerian numbers. Finally, we provide Wolfram Mathematica programs for the efficient verification of the derived identities.

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Mathematica Programs

The Mathematica programs in the project repository validate the results listed below.

Mathematica Program	**Result Validated**
`ValidateOrdinarySumsOfPowers.txt`	Proposition (3.3)
`ValidateDoubleSumsOfPowers.txt`	Proposition (3.5)
`ValidateMultifoldSumsOfPowers.txt`	Theorem (3.7)
`ValidateMultifoldBinomialSumsOfPowers.txt`	Proposition (3.8)
`ValidateOrdinarySumsOfPowersAtZero.txt`	Proposition (4.5)
`ValidateDoubleSumsOfPowersAtZero.txt`	Proposition (4.6)
`ValidateMultifoldSumsOfPowersAtZero.txt`	Theorem (4.7)
`ValidateMultifoldSumsOfPowersAtZeroBinomialForm.txt`	Proposition (4.8)

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